

大模型推理系统报告

Self-Attention 机制实现

实现这个最主要的挑战就是处理多头注意力机制，特别是当 Query 头数与 Key-Value 头数不一致时的情况。对于分组查询注意力(GQA)，将 Q 头和 KV 头按关系映射，使每组 Q 头对应一个 KV 头。计算时，通过嵌套循环遍历每个 KV 头及其对应的 Q 头组，计算它们之间的注意力分数。

文本生成

这个功能大部分已经实现，只需要根据提示以及自己的掌握的一些知识去实现，下面是整个功能的实现过程：

首先，处理用户输入的初始 prompt，将其传入模型进行前向计算。这一步骤包括：

- ✓ token 编码：将文本转换为模型可理解的 token ID
- ✓ 前向传播：通过模型的多层 Transformer 结构处理输入
- ✓ logits 计算：输出层生成词表大小的概率分布

然后，实现了 token 采样策略，包括：

- Temperature 调节：控制分布的"平滑度"，影响生成文本的随机性
- Top-K 采样：只从概率最高的 K 个 token 中选择
- Top-P 采样：选择概率累加达到阈值 P 的最小 token 集合

采样得到新 token 后，将其作为下一轮的输入，并更新 KV 缓存。这个过程循环进行，直到达到指定的最大长度或生成结束标记。为提高生成质量，实现了动态 temperature 调整机制，在检测到重复内容时提高 temperature 值，增加随机性，避免模型陷入重复生成的循环。

```
Welcome to My Rust4LLM Console

Available modes:
1. Story - Creative writing mode
2. Chat - Interactive conversation mode
3. API - Start API service
4. Exit

Please select a mode (1-4): 1

Loading story model...
Story mode initialized! Type 'exit' to return to main menu.

Story beginning: yesterday, i go to school

Generating story continuation...

yesterday, i go to school. She had to be eager to do a new thing: "I go with my big tea parts and a new shoe. But no one day she had to do was over, faster and faster. She thought it was time for it.
One day she saw something strange in the ground comfortable and she was very scared. She ran over to the ground and put the glove and a few egg in her hand. She felt brave and sad.
The stranger came back in and saw her and shouted back. She had made her combs singing and it ran and ran away from them. She felt sad and scared, but no one was gone, just how the ground again.
The one had made a loud noise in Bossy wasn't boring anymore. He knew his mom and dad were also brave. He was so sad and responded by himself. He knew he had been brave!
The braceowl sound and the computer and the ground was coming from his again. Everyone was so pleased with them and they became friends and they both cheered.<|end_story|>

Story beginning: █
```

图 1.故事续写功能

AI 对话

聊天功能是对文本生成的进一步扩展。我实现了支持多轮对话的系统，关键点包括：

对话历史管理：

- 创建数据结构记录对话消息，包括角色(用户/助手)和内容
- 实现添加新消息和清除历史的功能
- 设计历史长度限制机制，防止上下文窗口溢出

模板处理：

- 实现对话模板格式化功能
- 处理特殊标记<|im_start|>和<|im_end|>

响应生成与处理：

- ✓ 结合对话历史生成合适的 prompt
- ✓ 控制生成参数，如长度限制、温度等，使回复更自然
- ✓ 从生成结果中准确提取助手回复，过滤掉模板标记

会话状态维护：

- ◆ 实现对话过程中的上下文保持
- ◆ 系统消息设置，控制助手的行为风格
- ◆ 会话持久化，支持对话的暂停和恢复

```
Welcome to My Rust4LLM Console

Available modes:
1. Story - Creative writing mode
2. Chat - Interactive conversation mode
3. API - Start API service
4. Exit

Please select a mode (1-4): 2

📦 Loading chat model...
🌟 Chat mode initialized! Type 'exit' to return to main menu.
💡 Type 'clear' to clear chat history.

You: hello,who are you?
Assistant: I'm not capable of trying to talk about any other kind of responses. It's a good idea to ask a professional or professional person to ask about their experiences and concerns.

总生成 token 数: 38
总序列长度: 84

You: can you talk with me?
Assistant: You're welcome! I am here to assist you.

总生成 token 数: 13
总序列长度: 116
```

图 2.多轮对话

从图 2 可以看到，第二轮的虽然生成的 token 数明显少于第一轮，但是总序列长度是大于第一轮的，这是因为保留了第一轮的对话。

```
Please select a mode (1-4): 2

📦 Loading chat model...
🌟 Chat mode initialized! Type 'exit' to return to main menu.
💡 Type 'clear' to clear chat history.

You: hello,who are you?
Assistant: I'm not capable of trying to talk about any other kind of responses. It's a good idea to ask a professional or professional person to ask about their experiences and concerns.

总生成 token 数: 38
总序列长度: 84

You: can you talk with me?
Assistant: You're welcome! I am here to assist you.

总生成 token 数: 13
总序列长度: 116

You: clear
🌟 Chat history cleared!

You: hello
Assistant: I'm not sure, but I can provide you with a quick summary of the concept of quantum mechanics. Can you please provide me with the information you have about quantum mechanics?

总生成 token 数: 37
总序列长度: 73
```

图 3. 聊天记录清除功能

从图 3 可以看到，当我们输入 clear 之后再对话，之前的聊天记录已经清除了，这个可以从总序列长度看出来。

在实现 AI 对话功能时，也遇到了一些情况，比如说一个 token 重复多次或者一直输出的是换行，这个原因是因为两个模型的一些参数是不一样的，需要在 params.rs 中进行一定的修改，才能正常的运行。

API 实现

实现了基于 Actix-web 的 HTTP API 服务。主要考虑点包括：

服务架构：

- 采用异步处理模型，提高并发能力(这个在 **cpu** 上运行，作用并不大)
 - 实现惰性加载机制，仅在收到请求时才加载模型，减少资源占用
- 使用互斥锁保护模型访问，确保线程安全

API 端点设计：

- ✓ 状态检查接口：监控服务运行状态
- ✓ 故事生成接口：接收文本开头，返回完整故事
- ✓ 聊天对话接口：处理用户消息，维护对话上下文，返回助手回复

跨域与安全：

- 实现 **CORS** 支持，允许前端应用安全访问
- 请求参数验证，防止不合法输入
- 错误处理机制，提供友好的错误信息

```
You: clear
• Chat history cleared!

You: hello
Assistant: I'm not sure, but I can provide you with a quick summary of the concept of quantum mechanics. Can you please provide me with the information you have about quantum mechanics?

总生成 token 数: 37
总序列长度: 73

You: exit

👋 Welcome to My Rust4LLM Console

Available modes:
1. Story - Creative writing mode
2. Chat - Interactive conversation mode
3. API - Start API service
4. Exit

Please select a mode (1-4): 3

🔵 Starting API service...
[2025-03-19T07:51:25Z INFO] learning_lm_rust::api] 启动API服务在 http://127.0.0.1:8000
服务器启动在 http://127.0.0.1:8000
[2025-03-19T07:51:25Z INFO] actix_server::builder] starting 32 workers
[2025-03-19T07:51:25Z INFO] actix_server::server] Actix runtime found; starting in Actix runtime
[2025-03-19T07:51:25Z INFO] actix_server::server] starting service: "actix-web-service-127.0.0.1:8000", workers: 32, listening on: 127.0.0.1:8000
```

图 4.启动 API 服务

如图 4 所示，在 127.0.0.1:8000 启动了 API 服务，打开终端，输入命令测试续写故事功能和 AI 对话功能。


```
Last login: Wed Mar 19 11:44:24 on ttys000
(base) ➔ ~ curl -X POST http://127.0.0.1:8000/story \
-H "Content-Type: application/json" \
-d '{"prompt": "once a time"}'
{"status": "success", "data": ". It was old and had a pet bird on it. It had many p
owder that it looked so shiny and it could not. All the petal ropes in its fur t
o it on and it broke.\nOne day, it saw something incredible. It had been playing
with it. It made the sock. It had been not just a toy. The poor had many colors
, but it did not look in his. The chest was too heavy for Amy who could not reac
h it.\n\nThe bird had an idea. It tried to pull the lion to the bird, but then it
started to move. It fell from the heavy powder it far away. Amy and the bird wer
e sad.\n\nFinally, the bird came up to the doctor's hand. It made the bird free ou
t. The bird had rooded its neck and rode the bird flew far away. The bird was ver
y happy and thanked the bird and went back to its home. The bird was not scared
anymore and did not need anymore. The lion was sad because he could not make the
bird strong.<end_story>"}
```

```
Starting API service...
[2025-03-19T07:51:25Z INFO learning_lm_rust::api] 启动API服务在 http://127.0.0.1:8000
[2025-03-19T07:51:25Z INFO actix_server::builder] starting 32 workers
[2025-03-19T07:51:25Z INFO actix_server::server] Actix runtime found; starting in Actix runtime
[2025-03-19T07:51:25Z INFO actix_server::server] starting service: "actix-web-service-127.0.0.1:8000", workers: 32, listening o
n: 127.0.0.1:8000
[2025-03-19T07:56:21Z INFO actix_web::middleware::logger] 127.0.0.1 "GET / HTTP/1.1" 404 0 "-" "Mozilla/5.0 (Macintosh; Intel M
ac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36" 0.000080
[2025-03-19T07:56:23Z INFO actix_web::middleware::logger] 127.0.0.1 "GET / HTTP/1.1" 404 0 "-" "Mozilla/5.0 (Macintosh; Intel M
ac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36" 0.000091
[2025-03-19T07:56:24Z INFO actix_web::middleware::logger] 127.0.0.1 "GET / HTTP/1.1" 404 0 "-" "Mozilla/5.0 (Macintosh; Intel M
ac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36" 0.000087
[2025-03-19T07:56:25Z INFO actix_web::middleware::logger] 127.0.0.1 "GET / HTTP/1.1" 404 0 "-" "Mozilla/5.0 (Macintosh; Intel M
ac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/134.0.0.0 Safari/537.36" 0.000092
[2025-03-19T08:02:01Z INFO learning_lm_rust::api] 首次请求, 加载story模型...
[2025-03-19T08:02:02Z INFO actix_web::middleware::logger] 127.0.0.1 "POST /story HTTP/1.1" 200 908 "-" "curl/8.7.1" 0.979798
```

图 5.故事续写功能 API

```
Last login: Wed Mar 19 16:16:34 on ttys000
(base) ➔ ~ curl -X POST http://127.0.0.1:8000/chat \
-H "Content-Type: application/json" \
-d '{"message": "how are you?"}'
{"status": "success", "data": "I am not capable of using AI or other technical tool
s."}
(base) ➔ ~ curl -X POST http://127.0.0.1:8000/chat \
-H "Content-Type: application/json" \
-d '{"message": "hi"}'
{"status": "success", "data": "I am not able to perform such a task, but it's possi
ble."}
(base) ➔ ~
```

```
总生成 token 数: 13
总序列长度: 52
[2025-03-19T08:18:56Z INFO actix_web::middleware::logger] 127.0.0.1 "POST /chat HTTP/1.1" 200 84 "-" "curl/8.7.1" 5
.758943
I am not able to perform such a task, but it's possible.
总生成 token 数: 17
总序列长度: 83
[2025-03-19T08:19:34Z INFO actix_web::middleware::logger] 127.0.0.1 "POST /chat HTTP/1.1" 200 86 "-" "curl/8.7.1" 9
.498823
```

图 6.AI 对话 API

以上就是整个系统的实现。目前还在学习 CUDA, 能够实现矩阵相乘的算子, 原计划是想使用 CUDA 算子进行优化, 但是和 rust 结合起来, 自己还无法做到, 未来计划对这个项目进行优化:

- 使用 CUDA 算子, 将一些计算密集型内容放在 GPU 上进行运算。
- 此次模型效果并不好, 未来尝试使用比较流行的模型, 结合 rust 实现一个体验效果好的系统
- 对混合精度推理不是很了解, 可以尝试实现这个, 完善整个项目。